

CONTACT INFORMATION	Oxford, MS 38655 gavin.s.davies@gmail.com · gsdavies@olemiss.edu	(+1) 331-442-3845 https://gavinsdavies.com
BACKGROUND	Experimental High Energy physicist studying neutrino oscillations, with leadership roles in the software and computing infrastructure underpinning large-scale international collaborations	
PROFESSIONAL EXPERIENCE	Associate Professor of Physics & Astronomy University of Mississippi , Dept. of Physics & Astronomy, Oxford, MS USA University of Mississippi Center for Multi-Messenger Astrophysics Affiliate	July 2025 - Present
	Assistant Professor of Physics & Astronomy University of Mississippi , Dept. of Physics & Astronomy, Oxford, MS USA University of Mississippi Center for Multi-Messenger Astrophysics Affiliate	Aug. 2019 - July 2025
	Postdoctoral Research Fellow Indiana University , Dept. of Physics, Bloomington, IN USA Supervisor: Prof. Mark Messier	Oct. 2014 - Aug. 2019
	Postdoctoral Researcher Iowa State University , Dept. of Physics, Ames, IA USA Supervisor: Prof. Mayly Sanchez	Jul. 2010 - Oct. 2014
	Doctor of Philosophy (Ph.D.) in Physics Lancaster University , Dept. of Physics, Lancaster, UK Dissertation: 'Understanding and calibrating the DS-ECal of ND280 in the T2K project' Ph.D. Advisor: Dr. Laura Kormos	Oct. 2006 - July 2010
EDUCATION	M.Phys (Hons.) Physics / USA Lancaster University , Dept. of Physics, Lancaster, UK Dissertation: 'A Study of Charge Transfer Inefficiency in a Simulated CCD Damaged by Particle Radiation'	Oct. 2002 - July 2006
	USA Study Abroad Program, Michigan State University , MI USA	Aug. 2004 - May 2005
LANGUAGES	English (native) · Spanish (B1) · French (A2) · Danish (A1)	
LEADERSHIP EXPERIENCE	DUNE Phlex Adoption Working Group co-lead	May. 2026 - Present
	DUNE Global Software Technical Lead	Jan. 2026 - Present
	EMPHATIC Software & Analysis Coordinator	Oct. 2022 - Present
	NOvA Institutional Board Chairperson Elected position	Mar. 2021 - Mar. 2023
	NOvA Computing Coordinator	Jan. 2020 - Present
	Fermilab Users Executive Committee Chairperson Elected position	Sep. 2018 - Sep. 2019

	NOvA Exotic Neutrino Oscillation Analysis Convener	Oct. 2017 - Sep. 2022
	NOvA Production convener, Software Framework manager	Feb. 2011 - July 2014
COMMITTEES, SERVICE, AND AWARDS	APS DPF Coordinating Panel for Software & Computing [CPSC] Panel member, Rising Star Awards Sub-Committee Lead	Mar. 2025 - Present
	UM Excellence in Community Engaged Teaching and Learning Award Team member, Dept. of Physics & Astronomy Astronomy Art Contest	2024, 2025
	International Conference of High Energy Physics [ICHEP24] Computing co-chair, Prague	July 2024
	Graduate Program Coordinator, Dept. of Physics & Astronomy, University of Mississippi	Jan. 2025 - Present
	APS Formation Task Force for the Coordinating Panel for Software and Computing	Dec. 2023 - July 2024
	APS DPF Particle Physics Planning: Topical Group convener End-User Analysis	Apr. 2020 - July 2023
	Department of Energy Office of Science User Facilities Roundtable Committee member	Oct. 2020 - June 2021
	Flavor Physics and CP Violation (FPCP) international conference Organizing Committee	May 2021 - May 2022
	NOvA Executive Committee Elected position	May 2018 - Present
	Advocacy for High-Energy Physics Funding UEC Washington D.C. Trip	2017 - Present
	Fermilab Users Executive Committee (UEC) Elected position	Sep. 2017 - Sep. 2019
	Breakthrough Prize in Fundamental Physics	2016
AFFILIATIONS	APS DPF · SESAPS · NOvA Collaboration · EMPHATIC Collaboration · DUNE Collaboration	
OUTREACH & PUBLIC ENGAGEMENT	Dept. of Physics & Astronomy Astronomy Art Contest Judge	2024, 2025
	Mississippi Region VII Science Fair Judge	2025

University of Mississippi Academic Quiz Tournament Judge	2020, 2021, 2023, 2024
8th Annual Mississippi FIRST Tech Challenge Robotics Championship Judge	Feb. 2020
UM Spooky Physics Annual Halloween participant (Mr. Freeze demo)	2022 - Present
Oxford Science Cafe Quizmaster (during COVID-19 pandemic)	2020
Graduate Fair Informational Tables Dept. of Physics & Astronomy representative	various, 2021 - Present

GRANTS

External Funding; Total Awarded: \$4,576,992

Establishing a Scientific Computing Hub at the University of Mississippi Sep. 2025 – Aug. 2027
Principal Investigator, U.S. Dept. of Energy (EPSCoR), Award DE-SC0026215, \$1,800,000
Co-PIs: B. Quinn, J. Bennett (U. Mississippi), S. Tu, S. Yang (Jackson State U.)

LBNF/DUNE-US Operations Apr. 2026 – Mar. 2027
Principal Investigator, Fermilab, Award #734754, \$13,349

URA Visiting Scholars Program: EMPHATICally Preparing for DUNE Jun. 2026 – Nov. 2026
Principal Investigator, Universities Research Association, Award #2026-S-24, \$17,000

URA Visiting Scholars Program: EMPHATICally PINning DUNE Jun. 2025 – May 2026
Principal Investigator, Universities Research Association, Award #2025-S-13, \$16,000

LBNF/DUNE-US Operations Oct. 2025 – Mar. 2026
Principal Investigator, Fermilab, \$27,143

Intensity Frontier Studies at the University of Mississippi Sep. 2023 – Mar. 2026
co-Principal Investigator, U.S. Dept. of Energy (HEP), \$1,895,000
Co-PIs: B. Quinn, J. Bennett (U. Mississippi)

Intensity Frontier Studies at the University of Mississippi Nov. 2020 – Apr. 2023
co-Principal Investigator, U.S. Dept. of Energy (HEP), Award DE-SC0021616, \$800,000
Co-PI: B. Quinn (U. Mississippi)

Stimulating and Supporting Interest in STEM Fields with Hands-On Masterclass Programs May 2025 – Dec. 2026
co-Principal Investigator, University of Mississippi Mississippi Impact Grant, Account 250223370A, \$8,500
PI: J. Bennett (U. Mississippi)

Travel Awards

ORSP Travel Grant

May 2026

Travel to CHEP 2026 (Bangkok, Thailand), parallel talk, U. Mississippi ORED, \$1,200

ORSP Travel Grant

Oct. 2024

Travel to NBI 2024 (Mito, Japan), invited talk, U. Mississippi ORED, \$1,200

ORSP Travel Grant

Jul. 2024

Travel to ICHEP 2024 (Prague, Czechia), computing track convenor, U. Mississippi ORED, \$1,188

Pending / Under Review; Total Pending: \$4,162,950

Intensity Frontier Studies at the University of Mississippi

Apr. 2026 – Mar. 2029

co-Principal Investigator, U.S. Dept. of Energy (HEP), \$3,076,482 requested

Co-PIs: B. Quinn, J. Bennett (U. Mississippi)

FEC: From Fundamental Physics to Real-World Applications: Training the Next Generation in AI/ML and Image Recognition

Sep. 2026 – Aug. 2030

co-Principal Investigator, NSF EPSCoR (Research Infrastructure Improvement: Focused EPSCoR Collaborations), \$861,066 (UM share of \$5,865,499 total)

Multi-jurisdiction proposal: Wichita State U. (lead), U. Kansas, Kansas State U., U. Nebraska, U. Mississippi

AI-Enhanced DUNE Framework and Simulation Performance

Jul. 2026 – Mar. 2027

Collaborator, DOE Genesis Mission subaward (PNNL lead), \$112,701 (UM share of \$750,000 total)

Phlex as a Trustworthy AI Framework

Jul. 2026 – Mar. 2027

Collaborator, DOE Genesis Mission subaward (Fermilab lead), \$112,701 (UM share of \$750,000 total)

TEACHING/STUDENT SUPERVISION

Teaching

- PHYS 211, 212 Physics for Scientists and Engineers I/II
- PHYS 709 Advanced Mechanics (graduate-level)
- PHYS 503 Science Communication (graduate-level)
- PHYS 510 Department Colloquium

Postdoctoral Researchers

- Daniel Souza Correia, University of Mississippi
DUNE 2026 - Present
- Jeffrey Kleykamp, University of Mississippi
NOvA / DUNE 2021 - Present

Ph.D. Students

- Noah Knutson, University of Mississippi
EMPHATIC / DUNE 2024 - Present
- Xiaoyan Huang, University of Mississippi
DUNE / NOvA 2023 - Present
- Bishnu Acharya, University of Mississippi
NOvA 2021 - Present
- Andrew Dye, University of Mississippi
Ph.D. Physics (NOvA) 2021 - May 2025

- Now postdoctoral researcher at Wichita State University
- Luiz Ricardo Prais, University of Mississippi 2021 - Nov. 2024
Ph.D. Physics (NOvA)
Now postdoctoral researcher at University of Cincinnati

M.Sc. Students

- Ehsan Farooq, University of Mississippi 2024 - July 2025
M.Sc. Physics (DUNE)
Now pursuing Biomedical Engineering Ph.D. at University of Mississippi
- Devesh Bhattarai, University of Mississippi 2020 - Aug. 2023
M.Sc. Physics (NOvA / DUNE)
Now pursuing Physics Ph.D. at Mississippi State University

Undergraduate Students

- Suyog Badal, University of Mississippi Summer 2025 - Present
NOvA / DUNE, undergraduate research
- Samantha Reyes, University of Mississippi Summer 2026 - Present
MicroBooNE / DUNE machine learning, undergraduate research
- Chad Byrd, University of Mississippi Summer 2026 - Present
MicroBooNE / DUNE machine learning, undergraduate research
- Kaylyn Beard, University of Mississippi Summer 23, Fall 24
NOvA, undergraduate research
Now pursuing Physics Ph.D. at University of Minnesota

Additional Thesis Committee Service

- Served on 21 additional undergraduate honors, M.Sc., and Ph.D. thesis and prospectus committees since 2019, spanning Belle II, LIGO, Muon g-2, and theory in addition to NOvA/DUNE

EPIC (Exploring particle Physics Integrated with Computing) Summer School

- Coordinator and mentor, undergraduate summer training programme Jan. 2026 - Present
Mississippi EPSCoR Scientific Hub (MESH); recruitment, curriculum development, and student mentorship

Pedagogical Development

- ARISE@UM mentor
- CETL workshop, ‘Energizing and Motivating Students to Learn in Uncertain and Challenging Times’
- CETL workshop, ‘Top 5 Teaching Tips from Recent Pedagogical Research’

SELECTED PRESENTATIONS

June 2026 ‘New directions for joint neutrino oscillation measurements with T2K and NOvA’ (poster)
Neutrino 2026, University of California, Irvine, CA

June 2026 ‘Probing Non-Standard Interactions in NOvA with Bayesian MCMC methods’ (poster)
Neutrino 2026, University of California, Irvine, CA

May 2026 ‘NOvA Software and Computing: Evolution, Lessons, and Impact; Supporting a Decade of Physics’ (contributed talk)
CHEP 2026, Bangkok, Thailand

Apr. 2026 ‘Ten Years of NOvA in the Current Long-Baseline Picture, with DUNE on the Horizon’ (invited seminar)
Niels Bohr Institute, University of Copenhagen, Copenhagen, Denmark

Dec. 2024 ‘Exploring Exotic Neutrino Oscillations with the NOvA Experiment’ (invited plenary talk)

Miami 2024, University of Miami, Fort Lauderdale, FL

Nov. 2024 ‘Long-baseline neutrino experiments: Today and tomorrow’ (invited parallel talk)

SESAPS 2024, UNC Charlotte, North Carolina

Oct. 2024 ‘EMPHATIC’ (invited plenary talk)

NBI 2024, Neutrino Beams and Instrumentation, Tokai, Japan

Oct. 2024 ‘Hadron production measurements at EMPHATIC’ (invited parallel talk)

J-PARC symposium, Mito, Japan

Dec. 2023 ‘Long-baseline neutrino experiments: hoy y mañana’ (invited plenary talk)

8th ComHEP: Colombian Meeting on High Energy Physics, Ibagué, Colombia

July 2022 ‘Measurement of Standard and Non-standard Oscillations at NOvA’ (invited plenary talk)

XLI International Conference on High Energy Physics [ICHEP 2022], Bologna, Italy

Nov. 2021 ‘Latest oscillation results from NOvA’ (invited parallel talk)

SESAPS 2021, Tallahassee, Florida

Nov. 2019 ‘Latest three-flavor neutrino oscillation results from NOvA’ (invited parallel talk)

SESAPS 2019, Wrightsville Beach, North Carolina

Nov. 2019 ‘The Fermilab Users Executive Committee Model’ (invited parallel talk)

Society for Science at User Research Facilities (SSURF) annual meeting, College Park

Oct. 2019 ‘NOvA Software Infrastructure’ (invited parallel talk)

Inaugural Dark-matter and Neutrino Computation Explored (DANCE) workshop, Houston, TX

Aug. 2019 ‘Latest three-flavor neutrino oscillation results from NOvA’ (invited parallel talk)

Lepton Photon 2019, Toronto, Canada

Mar. 2019 ‘NOvA and T2K: Status and Outlook’ (invited plenary talk)

Aspen Winter Conference, Aspen, Colorado

Aug. 2018 ‘Neutrinos, antineutrinos, and deep learning on NOvA’ (invited seminar)

G.S. Davies, Seminar talk at Wichita State University, Wichita, KS

Sep. 2017 ‘Results and Prospects from NOvA’ (invited plenary talk)

19th International Workshop on Neutrinos from Accelerators (NuFACT 2017), Uppsala, Sweden

Aug. 2016 ‘Searches for Sterile Neutrinos with NOvA’ (invited parallel talk)

38th International Conference on High Energy Physics (ICHEP 2016), Chicago, Illinois, USA

Aug. 2016 ‘First Results from Searches for Active to Sterile Neutrino Oscillations with NOvA’

Fermilab JETP (Wine & Cheese) Seminar

